

Selective Entry and Structured Electricity-Recovery Performance in Japan’s Waste-Incineration Fleet: A Facility-Level Panel Study

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Abstract

Japan relies heavily on municipal waste incineration, but only 415 of 991 facilities (41.9%) generated electricity in fiscal year (FY) 2024 according to the official national summary. Fleet averages blur entry into electricity recovery and performance among facilities that already generate. Using Ministry of the Environment facility records for FY2005–FY2024, this paper examines both margins in one linked national panel. In a broad coded-asset risk set, larger prior-year capacity predicts first reporting of positive installed generation capacity and facilities older than ten years have lower adjusted entry probabilities than the 0–10-year reference group. Requiring positive prior-year throughput leaves the scale association intact but substantially attenuates the age gradient. This matters because 40 of 98 exact-year events follow zero or missing prior-year throughput and may represent commissioning, rebuild, or inactive-asset pathways rather than conversion of an operating plant. Among identifiable generators, year- and technology-adjusted gross electricity generation per tonne is lower at older-vintage facilities and higher at larger, more fully utilized facilities. Adjacent-year performance ranks correlate at 0.93, while entrants occupy approximately the middle of the same-year generator distribution during their first four observed years. These are descriptive associations, not causal effects of retrofit, aging, or entry. Asset entry, active-plant conversion, administrative attrition, and post-entry performance should not be managed as one average outcome.

Keywords: waste incineration; waste-to-energy; Japan; energy recovery; facility panel; transition

1 Introduction

Japan operates one of the world’s most incineration-dependent municipal waste systems, yet many facilities still burn waste without generating electricity from the heat they produce (Ministry of the Environment Japan, 2026; Uno, 2015; Tabata & Tsai, 2016; Sakai et al., 2011). In FY2024, the official national summary reports that 415 of 991 incineration facilities, or 41.9%, generated electricity (Ministry of the Environment Japan, 2026). Most facilities therefore remained outside electricity generation. This matters because Japanese municipal waste governance has long relied on thermal treatment for hygienic disposal, volume reduction, and local waste autonomy, while limited landfill space and strict environmental controls encouraged sophisticated incineration infrastructure (Brunner & Rechberger, 2015; Sakai et al., 2011). Sectoral planning now treats the electricity-generation gap as part of the waste sector’s decarbonization challenge (Yamada et al., 2023; Greenhouse Gas Inventory Office of Japan & Ministry of the Environment Japan, 2024). The

transition problem is therefore not whether Japan uses incineration. It is how observed entry into electricity recovery is patterned and how electricity recovery varies once generation is present.

Aggregate fleet summaries flatten two distinct questions. One is extensive: which facilities first report installed power-generation capacity? The other is intensive: among plants that already generate, which facilities achieve high electricity recovery per tonne? The system can therefore look uniformly slow in the mean while combining selective entry at one end of the fleet with persistent performance hierarchy at the other.

Research questions

This paper asks two descriptive empirical questions and one interpretive synthesis question. In the list, RQ denotes research question; the synthesis item states how the two empirical results are read together:

RQ1: Entry margin. Among coded facilities first observed without installed power-generation capacity, which prior-year age and capacity profiles are associated with first reporting positive capacity in the following fiscal year, and how do those associations change when the risk set is restricted to facilities with positive prior-year throughput?

RQ2: Generator-performance margin. Among identifiable operating generators, how is gross electricity generated per tonne associated with facility age/vintage, design capacity, utilization, heating value, and observed technology configuration within common fiscal years?

Synthesis. Taken together, do the observed entry pattern and generator-frame performance associations support one average-fleet modernization interpretation, or do they indicate distinct adoption and performance problems within the same incineration system?

The contribution is not a causal retrofit estimate or a technical optimization model. It is a linked facility-panel decomposition that separates broad coded-asset entry, conversion among demonstrably operating non-generators, and conditional generator performance. Scale selectivity appears in both entry frames, whereas the age gradient is pronounced in the broad frame but weaker in the active-conversion frame. Among generators, electricity recovery is structured by age/vintage, scale, and utilization. The same fleet can therefore combine different problems: asset entry, active conversion, and uneven performance after entry.

A natural expectation is that larger and newer facilities will have advantages. The analysis tests where that expectation appears, whether it survives changes in the risk-set definition, and whether entry is followed by convergence. The robust entry result is scale selectivity; the age result depends on whether the model describes all coded assets or operating non-generators. Entrants begin near the middle of the same-year generator distribution rather than at an obvious frontier. That sequence is what a fleet mean or generator-only study would miss.

The gap addressed here is narrower than a generic claim that Japan has been understudied. Waste-to-energy research can describe fleet trajectories, and generator-only studies can explain conditional performance once plants already operate as generators. What remains uncommon is one linked municipal-fleet analysis that estimates both margins and asks whether they point to the same modernization bottleneck. Japan makes that contrast visible because persistent non-generators, old small plants, and a more modern generating segment coexist in the same administrative system.

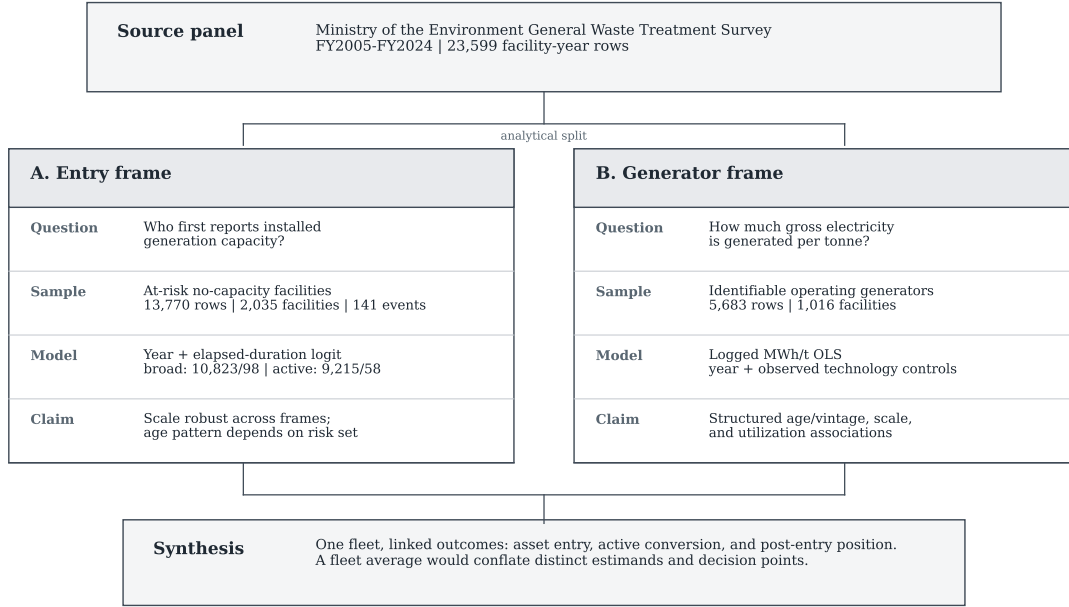


Figure 1: Analytical design separating the source panel into entry and generator-performance margins before interpreting both margins together.

In such mixed fleets, adoption and conditional performance should be modeled separately before they are interpreted together.

The rest of the paper proceeds as follows. Section 2 positions the paper against the literature most relevant to the analytical split. Section 3 introduces the two linked analytical frames and the main estimation choices. Section 4 reports the adoption and electricity-recovery results in sequence, then ties them together. Section 5 interprets the combined finding, explains what the data still cannot identify, and states a short set of evidence-consistent implications. Section 6 concludes.

2 Literature Positioning

This paper speaks to three literatures that often meet in practice but start from different empirical positions: waste-to-energy systems, facility-level performance analysis, and infrastructure lock-in. The review is organized around the paper’s central split between entry into generation and conditional performance after entry. The relevant gap is not simply that Japan needs another case study. It is that existing studies often describe fleets in aggregate or explain generator performance conditional on operation, yet rarely estimate both margins in one linked municipal-fleet design.

Work on waste-to-energy systems often documents national trajectories, technology choices, or lifecycle implications of thermal treatment (Astrup et al., 2009; Astrup et al., 2015; Brunner & Rechberger, 2015; Sun et al., 2018). Japan-specific work has emphasized technology upgrading, heat-use constraints, and decarbonization scenarios (Uno, 2015; Tabata & Tsai, 2016; Yamada et al., 2023). Sasao (2018) analyzes heat and electricity production at 635 Japanese waste-to-energy plants over FY2007–FY2015 using random-effects Tobit models and policy, technology, and

demographic correlates. Shino (2019) evaluates Japanese incinerator power-generation performance and supports electricity generation per unit waste input as an observable indicator while showing why thermal interpretation depends on calorific-value data. Neither follows annual first reporting from no installed capacity and reconnects that event to early post-entry generator position. Policy-facing work also treats waste-to-energy as useful only within a wider waste hierarchy and resource-recovery strategy (European Commission, 2017; Sakai et al., 2011).

Facility-level performance studies are closer to the present design, but they typically begin after entry has already occurred. Studies in Taiwan, for example, evaluate operating incinerators by decomposing waste-treatment, electricity-generation, or revenue performance within existing plants (Chen et al., 2012; Yeh, 2020). Other work focuses on energy-recovery criteria, plant scale, heat use, and the system consequences of different waste-to-energy configurations (Grosso et al., 2010; Münster & Meibom, 2010). Recent Chinese plant-level work is also highly informative about performance differentials inside the generating segment and the effectiveness of upgrading strategies at scale (Cui et al., 2026; Liu et al., 2025; Han et al., 2025). Those studies help define the intensive-margin question. They do not directly answer which non-generating facilities enter generation in the first place.

The present paper therefore uses those comparator papers as a method-positioning map rather than as templates to copy. The map below makes the adaptation logic explicit.

Comparator adaptation of research logic

Source family	Logic adapted for the Japan panel	Boundary of the adaptation
Cui et al. (2026)	Facility hierarchy motivates separate plant-level entry and performance analysis.	No optimization frontier or ranking on Cui et al.’s technical frontier.
Liu et al. (2025)	Effectiveness is separated from expansion: entry and post-entry MWh/t are different outcomes.	No China-style waste-energy-carbon development model.
Han et al. (2025)	Available furnace and operating-configuration controls enter the generator comparison.	No plant-level pollutant-control model.
Sasao (2018)	Japanese plant-panel comparison is retained, but the first outcome changes to annual installed-capacity entry.	No Tobit production function or identified policy mechanism.
Shino (2019)	Gross MWh/t is primary; heating-value-based conversion and reported efficiency provide checks.	Gross MWh/t is not net export or complete thermodynamic efficiency.
Chen et al. (2012) and Yeh (2020)	Operating incinerators are compared empirically after entry.	No data-envelopment-analysis frontier or revenue-inefficiency model.
Grosso et al. (2010) and Münster and Meibom (2010)	Electricity recovery is interpreted in a wider energy-system context.	No R1 calculation, lifecycle balance, or energy-system optimization.

This matters because the paper is not a weaker version of one comparator. Its foundation combines facility heterogeneity from incineration studies, transition timing from event-history methods, and conditional performance comparison from Japanese and international plant analyses. The paper changes the unit of inquiry from production among plants already observed with energy recovery to first observed entry among non-generators, then reconnects entrants to early generator performance. This is an adaptation of established methods to a different estimand, not a claim to have invented the estimators.

The lock-in literature adds a different expectation. Infrastructure performance may be shaped by durable design choices, inherited scale, and institutional arrangements rather than by frequent late-life reversals at mature facilities (Unruh, 2000; Geels, 2004; Seto et al., 2016). The useful implication here is not that incineration plants are literally irreversible. It is that cross-facility differences may matter more than repeated performance resets. In municipal infrastructure, that persistence can also be institutional rather than purely technical: plant networks are shaped by jurisdictional boundaries, merger histories, charging regimes, and the political difficulty of reorganizing service territories (Rausch, 2006; Sakai et al., 2008; Sakai et al., 2011). That expectation is only partially visible if entry into generation and performance within generation are never separated.

Read together, the literatures imply a practical identification problem for fleet studies. Fleet-level work can show aggregate progress but cannot tell whether low average performance reflects many non-generators, weak generator performance, or both. Generator-only work can estimate performance correlates after entry, but not who stays outside generation. Lock-in work explains why mature infrastructure may remain stratified, but not whether the key margin lies before or after entry. The contribution of this paper is to use one linked panel to expose those blind spots without treating them as one average process.

3 Data and Design

The analysis uses the Ministry of the Environment’s General Waste Treatment Survey for FY2005–FY2024 (Ministry of the Environment Japan, 2026; e-Stat, n.d.). From 23,599 facility-year rows, the coded frame retains 19,827 observations across 2,948 identifiable facilities. The survey contains generating and non-generating facilities in the same administrative system, allowing the paper to separate observed entry into installed capacity from performance after entry. The official FY2024 publication reports 415 electricity-generating facilities among 991 incineration facilities (41.9%). The reconstructed row-level panel has 417 of 1,014 FY2024 rows with positive installed capacity (41.1%). The official share is used for national context; the analytical share is only a reproducibility diagnostic because reconstruction and classification boundaries differ.

The design intentionally combines established empirical building blocks rather than inventing a new estimator. The adoption layer follows discrete-time event-history logic for a binary transition observed in annual administrative data (Allison, 1982; Beck et al., 1998). The generator layer follows facility-level performance studies that compare operating incinerators after entry, while using panel regressions to summarize persistent cross-facility structure (Chen et al., 2012; Yeh, 2020; Wooldridge, 2010). What is different here is the way those pieces are linked: the paper first models observed entry into generation, then separately models electricity recovered per tonne among identifiable generators, and only then asks whether a single average-fleet interpretation is adequate. This makes the framework similar in spirit to plant-level optimization and performance papers, but distinct in its Japan-specific two-margin diagnostic focus.

Several modeling choices follow from that alignment between question and sample. The adoption model is deliberately narrow because the event of interest is first observed reporting of generation, not every capital investment that may have occurred before or outside the panel. The generator model is also deliberately narrow because it compares identifiable operating generators, not all facilities and not uncoded generator rows that cannot be followed reliably as facility panels. The paper therefore uses the same administrative source for both questions but does not force both

questions into the same sample or the same estimator.

Research-question-to-model bridge			
Question	Main frame and model	What it can show	What it cannot show
RQ1: entry	Broad coded-asset risk set; active-operation restriction; exact one-year logit hazard with year indicators and elapsed time at risk	Which prior-year profiles predict entry and which associations survive positive prior throughput	A causal retrofit effect, verified first operating date, or full capital-history mechanism
RQ2: generator performance	Identifiable generators; year- and technology-adjusted OLS for logged bounded MWh/t	How performance differs across age/vintage, scale, utilization, heating value, and observed configuration	Net export, complete thermodynamic efficiency, facility fixed effects, or causal effects
Synthesis	Two linked but non-identical frames read together	Whether one average-fleet interpretation hides two different bottlenecks	A strict causal chain from adoption into generation to later generator performance

The first frame is the coded installed-capacity entry frame. A facility is classified as having generation capacity when reported installed power capacity is positive. Facilities first observed without that capacity remain at risk until they first report a positive value or until their last coded observation. After excluding 913 left-censored facilities already reporting positive capacity in their first observed year, the risk set contains 13,770 facility-years across 2,035 facilities, with 141 installed-capacity entry events.

That 141-event count is the descriptive adoption universe used for event-rate summaries and the pathway audit. The main hazard model below uses a stricter exact one-fiscal-year lagged subset with 98 retained events; the 42 timing-ambiguous non-adjacent events and 1 unresolved event remain in pathway and sensitivity evidence rather than being forced into the headline hazard.

In practical terms, the entry model asks whether a facility reporting no installed generation capacity in one fiscal year first reports positive capacity in the next. The exact one-fiscal-year lagged logit hazard contains 10,823 observations, 1,911 facilities, and 98 events. Predictors are prior-year age band and design capacity, fiscal-year indicators, and actual elapsed fiscal time since first entering the risk set. Facility-clustered standard errors account for repeated observations. Elapsed time is calculated from fiscal-year values rather than by counting observed rows: the two measures differ for 4,055 exact-lag observations because official-code gaps interrupt the panel. The model estimates 19 parameters, or 5.16 events per parameter. A year-plus-prefecture model is supplemental because it estimates 64 parameters with only 98 events. The exact-year restriction prevents the FY2010–FY2012 code gap from being treated as a one-year lag.

Formally, let $A_{it} = 1$ if facility i first reports positive installed generation capacity in fiscal year t , conditional on still being in the at-risk set $R_{it} = 1$. The main entry model is:

$$\Pr(A_{it} = 1 \mid R_{it} = 1) = \text{logit}^{-1} [\alpha + \beta_1 I(\text{Age}_{i,t-1} = 10\text{--}20) + \beta_2 I(\text{Age}_{i,t-1} = 20\text{--}30) + \beta_3 I(\text{Age}_{i,t-1} \geq 30) + \beta_4 \text{Capacity100}_{i,t-1} + \delta D_{it} + \gamma_t]. \quad (1)$$

Here, i indexes facilities, t fiscal years, and $I(\cdot)$ is an indicator. The omitted age group is 0–10 years. γ_t contains fiscal-year indicators, Capacity100 measures design capacity per 100 tonnes/day,

and D_{it} is actual elapsed at-risk duration in ten-fiscal-year units. Table 2 reports average marginal effects (AMEs) in percentage points (pp), not log-odds. The broad-frame -1.41 pp AME for age 10–20 is a model-averaged probability difference from the 0–10 reference; the +0.45 pp capacity AME is the average change per additional 100 t/day. Neither is a causal retrofit effect.

Age and capacity are observable pre-event profiles, not isolated mechanisms. A starting-year reset can make a rebuilt or recommissioned asset appear young, so prior-year rather than event-year profiles are used. The broad model is an asset-entry estimand: 40 of 98 exact-year events have zero or missing prior-year throughput. A required active-operation sensitivity therefore restricts the risk set to positive prior-year throughput. It contains 9,215 facility-years, 1,663 facilities, and 58 events with the same timing terms and clustered uncertainty. A prior-technology sensitivity, positive-output event definition, post-entry bridge, and panel-exit diagnostic further bound interpretation. Panel exit is not closure because identifier change, consolidation, and reporting loss remain possible.

The second frame is the canonical generator frame: operating facilities with positive throughput and positive output after standard cleaning and bounding. Of 6,660 operating-generation rows, 907 lack official facility codes and are excluded, leaving 5,683 observations across 1,016 facilities. The dependent variable is log gross megawatt-hours per tonne (MWh/t), clipped to 0.01–0.80 before logging. Main predictors are age, design capacity, capacity utilization, and heating value. The primary model also adjusts for normalized furnace type, operating mode, facility type, and number of furnaces. Utilization equals annual throughput divided by design capacity times 365 and is capped at 1.0. The outcome is gross electricity-recovery intensity, not net export, revenue, useful heat, or complete thermodynamic efficiency.

Let q_{it} be the clipped electricity recovery ratio and $y_{it} = \log(q_{it})$. The primary model is:

$$y_{it} = \alpha + X'_{it}\beta + Z'_{it}\theta + \gamma_t + \varepsilon_{it}, \quad (2)$$

where X_{it} contains age, design capacity per 100 t/day, capped utilization, and heating value; Z_{it} contains observed technology-configuration controls; and γ_t absorbs common fiscal-year conditions. Ordinary least squares (OLS) is used for a transparent year-adjusted cross-facility comparison, with facility-clustered standard errors (Wooldridge, 2010). Age is derived from fiscal year and reported starting year. With year indicators, its coefficient primarily contrasts facilities of different age and technology vintage in the same year; it does not identify the effect of making one plant older.

A supplemental estimator ladder reports pooled OLS, year-indicator OLS, random effects (RE), and year-indicator plus RE models. A within-between diagnostic separates facility means from deviations. These variants do not remove unobserved-facility concerns, and facility fixed effects would answer a different within-plant question while absorbing durable scale and vintage structure. As a persistence diagnostic, the analysis correlates exact adjacent-year within-year percentile ranks.

Because the outcome is logged, small coefficients approximate proportional differences. In the primary model, -0.0329 for age/vintage corresponds to about 3.2% lower gross MWh/t per additional age year in a cross-facility comparison. The +0.1103 capacity coefficient corresponds to about 11.7% higher gross MWh/t per 100 t/day using $100[\exp(0.1103) - 1]$. These are comparisons, not forecasts from changing a facility attribute.

The results apply to the canonical identifiable generator frame, not all generation activity. Uncoded operating rows are concentrated in FY2010–FY2012, so early/late comparisons are coded-window diagnostics, not a Fukushima identification design. The bounded MWh/t outcome places the paper

Table 1: Linked analytical framework

Margin	Linked sample	Empirical question	Paper role
Installed-capacity entry	Broad risk set: 13,770 facility-years and 141 events; exact model: 10,823 rows and 98 events	Which prior-year profiles predict broad coded-asset entry?	Identifies robust scale selectivity and a broad-frame age gradient
Active conversion	Positive-prior-throughput frame: 9,215 rows, 1,663 facilities, 58 events	Do associations persist among active non-generators?	Scale persists while age attenuates
Electricity-recovery margin	Canonical generator frame: 5,683 observations across 1,016 operating generators	How does gross electricity generated per tonne vary once generation already exists?	Shows whether mature generator performance remains structured
Synthesis	Two linked but non-identical analytical frames	Would one average-fleet view misstate the modernization bottleneck?	Shows why entry and mature performance should not be read as one average process

Note: entry uses lagged discrete-time hazards. Generator performance primarily uses year- and technology-adjusted OLS; pooled and random-effects (RE) variants are supplementary. Year indicators are not facility fixed effects.

closer to applied energy-recovery studies than plant optimization alone (Grosso et al., 2010; Münster & Meibom, 2010; Shino, 2019).

Administrative checks address duplicate official codes and plausible heating values. In a 4,971-row validation frame, year- and technology-adjusted models of a heating-value-based thermal-conversion proxy and reported generation efficiency preserve negative age/vintage and positive scale and utilization associations. The logged outcomes correlate at 0.8636 because they use related survey fields. An exact-adjacent-year model with lagged predictors also preserves the pattern. These are convergent checks, not independent experiments.

Generative AI tools were used during code development and review to suggest implementations, test cases, documentation structure, and reproducibility checks. The author executed the pipeline, inspected source records and generated outputs, checked reported quantities against machine-readable manifests, and retained responsibility for all definitions, model choices, interpretations, and revisions. The tools did not autonomously acquire, alter, or validate the underlying administrative data.

The layers answer complementary parts of one modernization problem. Entry analysis identifies which coded assets appear to enter generation and whether results change among active non-generators. Generator analysis then locates performance differences after entry. Without both, the paper cannot distinguish asset entry, active conversion, and post-entry position.

The design is diagnostic rather than structural. It models observed transition within stated risk sets and conditional generator performance within an identifiable frame. Variance and rank diagnostics describe the data structure; they do not resolve omitted-variable concerns or make any model causal.

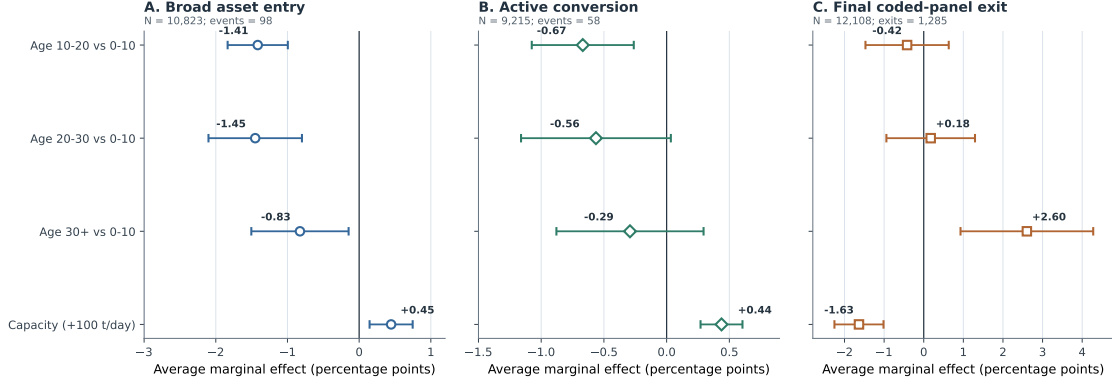


Figure 2: Average marginal effects and facility-clustered 95% confidence intervals for broad coded-asset entry, active conversion requiring positive prior-year throughput, and final coded-panel exit. Entry hazards include actual elapsed at-risk duration. Panel exit is not verified closure.

4 Results

4.1 Installed-capacity entry is selective rather than diffuse

The most stable entry result is scale selectivity. Exact-year event rates rise from 0.15% in the smallest prior-year capacity quartile to 2.49% in the largest, which contains 62 of 98 modeled events. Raw age rates are nonmonotonic: 1.97% at ages 0–10, 0.18% at 10–20, 0.72% at 20–30, and 1.68% at 30 or more. Events cluster in FY2013–FY2019, but this is not interpreted as an identified policy shock.

In the broad coded-asset model, adjusted age-band AMEs are -1.41, -1.45, and -0.83 percentage points (pp) relative to ages 0–10. Each additional 100 t/day predicts +0.45 pp annual entry probability. The model includes actual elapsed duration and has pseudo- $R^2 = 0.1920$. Requiring positive prior-year throughput leaves capacity at +0.44 pp but shrinks the age differences to -0.67, -0.56, and -0.29 pp; only the first is significant at 1%, while the latter two have $p = 0.064$ and $p = 0.329$. Entry is therefore consistently scale-selective, while the age gradient depends on whether the estimand includes commissioning, rebuild, or inactive-asset pathways.

Prior-technology adjustment, a positive-output event, saturated year-plus-prefecture models, broader lag definitions, and alternative capacity forms do not reverse the scale result. They do not identify the event mechanism.

The pathway audit bounds mechanism. Of 141 events, 50 are reset- or rebuild-like, 36 are continuity-type upgrades, 12 are forward-dated or placeholder entries, 42 are timing-ambiguous non-adjacent events, and 1 is unresolved. This mix does not identify replacement, refurbishment, new build, or reporting change as the unique mechanism.

Panel disappearance is a competing observed path. Among 1,894 facilities with no capacity event, 1,305 (68.9%) are last observed before FY2024. In a separate next-year hazard, age 30 or more has a +2.60 pp AME relative to age 0–10, while each 100 t/day of capacity has a -1.63 pp AME. The younger age bands are not statistically distinguishable from the reference. This is coded-panel exit, not verified physical closure.

Table 2: Lagged hazard results for broad entry, active conversion, and coded-panel exit

Variable	Broad entry AME (pp)	Active conversion AME (pp)	Panel exit AME (pp)
Prior-year age 10–20 years (versus 0–10)	-1.41 (0.21)	-0.67 (0.21)	-0.42 (0.54)
Prior-year age 20–30 years (versus 0–10)	-1.45 (0.33)	-0.56 (0.30)	0.18 (0.57)
Prior-year age 30+ years (versus 0–10)	-0.83 (0.35)	-0.29 (0.30)	2.60 (0.85)
Prior-year capacity (per 100 t/day)	0.45 (0.15)	0.44 (0.09)	-1.63 (0.32)
Observations	10,823	9,215	12,108
Facilities	1,911	1,663	2,022
Events	98	58	1,285

Note: entries are average marginal effects (AMEs) in percentage points with facility-clustered standard errors in parentheses. The first two columns use exact one-fiscal-year lags, fiscal-year indicators, and actual elapsed at-risk duration; active conversion requires positive prior-year throughput. Panel exit is an administrative outcome, not verified closure.

4.2 Electricity recovery within generation is strongly structured

Within the canonical regression frame, generators differ persistently from one another. The within-to-total variance ratio of pooled log electricity-recovery intensity is 0.1499, so most observed variation lies between facilities. More directly, 4,368 exact adjacent-year pairs across 915 facilities have a pooled within-year percentile-rank correlation of 0.9325; the median annual correlation is 0.9323, with a range of 0.8848–0.9763. A high-performing facility therefore usually remains high-ranked one year later even though its exact MWh/t can change. The variance ratio falls from 0.1795 in FY2005–FY2009 to 0.0956 in FY2013–FY2024. This is not a Fukushima identification design because FY2010–FY2012 generator rows lack official facility codes. These diagnostics show persistence, not causal isolation of vintage from technology, governance, maintenance, or other durable characteristics.

The primary model gives the same directional structure after common-year and observed-technology adjustment. The age/vintage coefficient is -0.0329, capacity is +0.1103 per 100 t/day, and utilization is +0.7600; heating value is not statistically distinguishable from zero. Adding furnace type, operating mode, facility type, and furnace count raises R^2 from 0.3699 to 0.3830 without materially changing the focal coefficients. These are conditional associations, consistent with facility-level evidence on uneven performance (Chen et al., 2012; Sasao, 2018; Shino, 2019; Yeh, 2020; Grosso et al., 2010), not intervention forecasts.

The electricity-recovery margin is structured rather than static. Facilities vary through utilization and operating conditions, but age/vintage and scale differences remain visible. Chinese studies show that gains can still follow technology, pollutant-control, classification, and load-rate improvements within generating systems (Liu et al., 2025; Han et al., 2025). The present rank evidence describes observed persistence; it does not bound attainable gains.

This is why the intensive margin cannot be inferred from the entry margin alone. A facility can re-report installed capacity and produce electricity while still occupying a different relative performance position from its peers. The data do not establish how much a specific operational intervention could close those gaps. They show structured and highly persistent performance rather than a simple continuation of capacity entry.

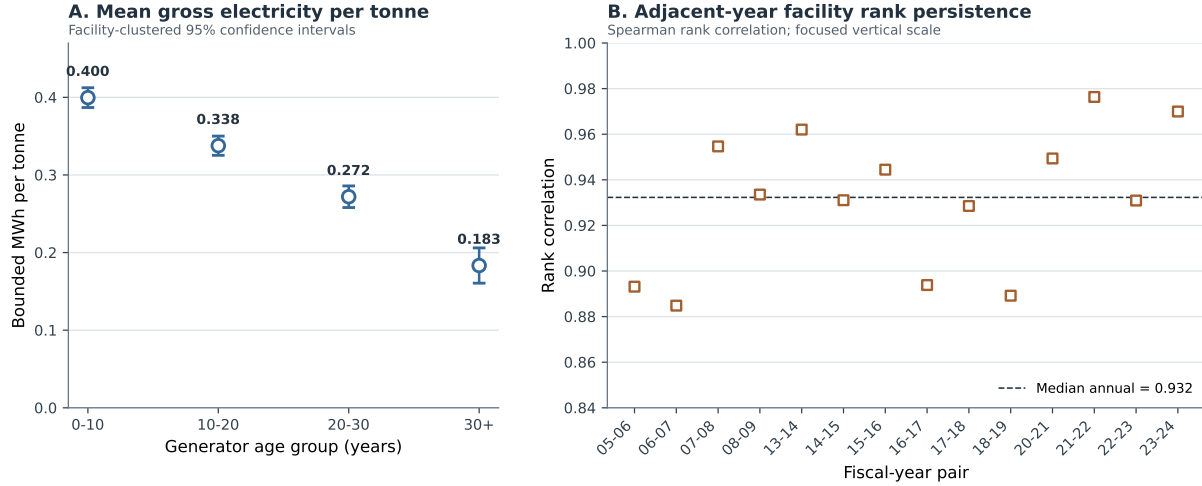


Figure 3: Mean bounded gross electricity generation per tonne by facility-age group with facility-clustered 95% confidence intervals, and adjacent-year within-year percentile-rank correlations.

Table 3: Primary electricity-recovery specifications in the canonical generator frame

Variable	Base year-adjusted OLS	Primary year + technology OLS
Facility age/vintage (years)	-0.0348 (0.0023)	-0.0329 (0.0023)
Capacity (100 t/day)	0.1051 (0.0087)	0.1103 (0.0101)
Capacity utilization	0.7760 (0.1351)	0.7600 (0.1319)
Heating value (MJ/kg)	0.0033 (0.0021)	0.0032 (0.0020)
Fiscal-year indicators	Yes	Yes
Technology controls	No	Yes
Observations	5,683	5,683
Facilities	1,016	1,016
R-squared	0.3699	0.3830

Note: OLS means ordinary least squares and MJ/kg means megajoules per kilogram. Technology controls are normalized furnace type, operating mode, facility type, and furnace count. Standard errors are clustered by facility. Age, capacity, and utilization have $p < 0.001$; heating value has $p > 0.10$. The estimator ladder is in the supplement.

For interpretation, this matters as much as the entry result. If the paper only documented selective entry, a reader could infer that the main task was simply to push more facilities into generation. If it only documented generator hierarchy, a reader could infer that non-generators were lagging versions of the same problem. The linked result makes both shortcuts inadequate.

In a 4,971-row plausible-value frame, technology-adjusted thermal-conversion and reported-efficiency models preserve negative age/vintage and positive capacity and utilization coefficients. Their logged outcomes correlate at 0.8636 because they share survey inputs. An exact-adjacent-year model with lagged predictors also preserves all three signs. These are convergent checks, not independent experiments.

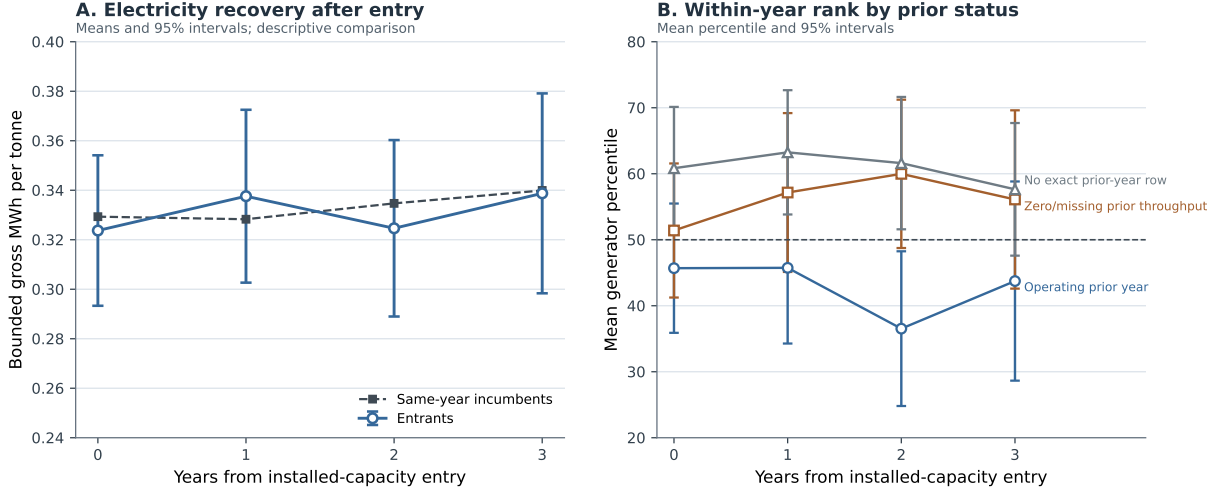


Figure 4: Early post-entry performance. Panel A compares entrants with same-year incumbent generators in gross MWh/t. Panel B reports mean within-year percentile by prior operating status. Points show means and bars show 95% confidence intervals; event counts decline over time.

4.3 Why the two results belong together

The entry analysis shows robust scale selectivity and a risk-set-dependent age pattern. The generator analysis shows durable cross-facility structure after entry. These margins do not form a strict causal chain, but a single fleet average would obscure whether an observation lies outside generation, is newly entering, or already generates.

Of 141 installed-capacity entrants, 137 appear in the canonical generator frame within three years, contributing 389 event-time observations. At event time zero, 125 entrants average 0.324 MWh/t versus 0.329 among same-year incumbents, with a mean within-year percentile of 51.5. An incumbent is a generator with no observed entry event or with an event in an earlier fiscal year. At event times one through three, mean percentiles are 54.8, 52.1, and 52.9; entrant-minus-incumbent differences are +0.009, -0.010, and -0.001 MWh/t. Entrants therefore appear near the middle of the contemporaneous distribution on average.

The trajectory is descriptive and selected: represented events decline from 125 to 71. Entrants operating in the prior year begin below the median rank, whereas those with zero/missing throughput or no exact prior row begin higher. These contrasts suggest that in-place conversion and commissioning/rebuild pathways are not interchangeable, but they are not causal estimates.

Only three entrants reverse the capacity flag in an observed next year, and 135 of 141 report positive output by the following year. Entry is usually followed by measurable operation, but neither the event nor trajectory proves that entry causes subsequent rank. Panel exit is a competing administrative path, not verified closure.

5 Discussion

The main interpretive claim is that broad asset entry, conversion of an operating non-generator, and performance after entry are not interchangeable outcomes. Larger prior-year capacity predicts entry

in both risk sets. The broad age gradient attenuates sharply among active non-generators because 40 of 98 exact-year events follow zero or missing throughput. Among generators, age/vintage, scale, and utilization remain associated with gross MWh/t after year and technology adjustment, while entrants begin near the middle of the same-year distribution.

The contribution is therefore not merely that new or large plants perform better. It tests that expectation against alternative populations and outcomes, finding a stable scale pattern, frame-sensitive age pattern, and no automatic post-entry frontier position. These boundaries provide a clearer foundation for later engineering and causal work than one fleet-average association.

This makes the paper useful as a planning diagnostic rather than as an intervention ranking. The practical value is triage: deciding which part of the fleet needs an entry-side asset question and which part needs a generator-side performance question. Table 4 translates the empirical patterns into defensible planning questions while keeping the claim boundaries explicit. In practical terms, the design gives planners a screening sequence: first decide whether a facility is an entry-side asset problem or an already-generating performance problem, then decide which more detailed engineering, governance, or capital-history review is worth pursuing.

Table 4: Planning diagnostic implied by the two-margin evidence

Empirical pattern	Planning use	Boundary condition
Scale-selective entry	Check whether small non-generators have comparable waste supply and system scale	Capacity is not a causal lever in this model
Age attenuates among active plants	Separate commissioning/rebuild from operating-plant conversion before using age to screen	Older active plants are not proven unable to convert
Mixed event pathways	Check asset histories before treating adoption as one retrofit mechanism	The audit does not distinguish replacement, refurbishment, and reporting change with certainty
Older/smaller panel exit	Verify closure, recoding, consolidation, or reporting status before interpreting continued non-entry	Administrative disappearance is not verified closure
Structured generator performance	Compare generators with similar age, scale, and utilization before judging improvement room	Coefficients are conditional associations, not causal effects
Entrants begin near the distribution middle	Evaluate entry and later operation as separate milestones	The trajectory is selected and does not estimate an entry effect

The interpretation has clear limits. The active-conversion model has only 58 events, and changing the risk set changes the estimand rather than formally testing coefficient equality. The pathway audit cannot identify a unique mechanism. Generator models omit municipal finance, contracting, maintenance quality, waste composition, and complete retrofit history. Installed capacity may not be usable capacity, gross output is not net export, useful heat is unmeasured, and panel exit is not closure. Follow-up falls from 125 entrants at time zero to 71 at time three. The defended claim is therefore scale-selective observed entry, a frame-dependent age association, and structured post-entry performance, not an intervention ranking.

The results do not identify the best intervention for a municipality. For non-generators, assessments should check scale, prior operation, and capital history before considering renewal, consolidation,

conversion, or continued non-generation. For generators, the separate question is whether operational or capital improvement is plausible within the observed age/vintage, scale, and utilization context.

That distinction is administrative as well as technical. Japanese waste systems are organized through municipalities whose planning boundaries do not always align with efficient waste sheds, and intermunicipal reorganization has its own political costs and institutional legacies (Rausch, 2006; Sakai et al., 2008). In that setting, collapsing non-generators and generators into one average segment risks hiding the difference between entry-side asset decisions, which are lumpy and governance-heavy, and generator performance assessment, which is more incremental and operational.

If entry and performance were one process, a fleet-average modernization story would be adequate. The evidence instead points to three diagnostic starting points: whether a record represents broad asset entry or active conversion, whether scale makes projects structurally comparable, and how performance should be evaluated after generation begins.

The analytical split may transfer beyond Japan, but the coefficients do not. It is most relevant where administrative data contain both facilities without energy-recovery equipment and repeated output records for generators. Replication elsewhere requires locally valid identifiers, capacity definitions, closure histories, electricity-use boundaries, and governance variables. Where nearly all plants generate or net heat and electricity exports are measured jointly, the relevant margins would differ.

The broader policy context points in the same direction. Comparative waste policy studies and European framework discussions both treat waste-to-energy as valuable only when embedded inside a wider hierarchy that preserves prevention, reuse, and recycling priorities (Sakai et al., 2011; European Commission, 2017). The empirical split helps explain why those hierarchies matter. A municipal system can have real energy-recovery gains available at the non-generating margin while still facing a different, narrower set of decisions inside the already-generating segment.

For climate interpretation, the relevant question is not simply whether a plant generates, but what kind of generator it is and how much improvement remains inside that segment. Waste-to-energy performance is judged against both an internal engineering standard and the emissions profile of the broader energy system, including the avoided-emissions logic built into carbon accounting (Astrup et al., 2009; Münster & Meibom, 2010). In Japan’s current decarbonization setting, where national inventories and scenario work track sectoral emissions and energy mix changes, the two-part design is useful because it keeps those questions separate (Greenhouse Gas Inventory Office of Japan & Ministry of the Environment Japan, 2024; Yamada et al., 2023).

That separation is also what makes the paper more understandable for a planning audience. Municipal systems rarely choose among abstract technological ideals. They decide whether to renew an aging non-generator, coordinate waste routing toward a larger plant, maintain an existing generator, or invest in an upgrade for a plant that already produces electricity. Those are related decisions, but they are not interchangeable. Keeping the extensive and intensive margins separate helps planners locate the bottleneck without relying on one blended fleet mean.

6 Conclusion

In the observed facility panel, first reporting of positive installed capacity is consistently scale-selective, but its age pattern depends on whether the risk set includes all coded assets or only facilities operating in the prior year. Final coded-panel exit is a competing administrative outcome. Among identifiable generators, gross MWh/t remains associated with age/vintage, scale, and utilization after common-year and observed-technology adjustment, and adjacent-year ranks are highly correlated. Most entrants soon report output and begin near the middle of the same-year generator distribution; this does not imply that entry causes rank. Fleet assessment should distinguish asset entry from active conversion and evaluate post-entry performance separately.

Acknowledgements

The author thanks Prof. Han Ji for supervision and critical feedback during the development of the underlying thesis project from which this paper is derived.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Contributor Roles Taxonomy (CRediT) Authorship Contribution Statement

Pann Phetra: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical Approval

This study uses publicly released administrative facility data and does not involve human participants, animal subjects, or private personal data.

Data Availability

The facility-level source data are derived from the publicly released Ministry of the Environment Japan General Waste Treatment Survey. Subject to source-file redistribution conditions, the author will deposit the analysis code, machine-readable result manifests, derived tables, figure-generation scripts, and manuscript figures in a versioned repository associated with the submitted paper. Raw workbooks can be obtained from the Ministry and e-Stat portals.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During preparation, the author used OpenAI Codex and Anthropic Claude for language revision, manuscript organization, and assistance with code development and review. The author executed the analyses, inspected source data and generated outputs, independently checked reported results, reviewed and edited all AI-assisted material, and takes full responsibility for the content. These tools were not used as authors and did not replace author judgment or accountability.

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